EDC Assignment#2

Last Date: 11-Oct-26; Provide the complete solution as a single Zip file

(student ID as file name)

From Transistors to Qubits: A Hybrid Learning Framework for Understanding Quantum Computing through Semiconductor Concepts:

Students possessing fundamental knowledge of semiconductor devices and transistor-based digital circuits often experience a conceptual gap when transitioning to quantum computing. This assignment proposes a transistor-to-qubit learning framework that maps familiar concepts such as transistor switching, binary states, logic gates and circuit operations to corresponding quantum concepts including qubits, quantum gates, superposition and measurement. A simulation-based experimental platform has to be developed using classical electronic circuits and a quantum computing simulator. The effectiveness of the framework will be evaluated through pre- and post-learning assessments, circuit implementation tasks and conceptual understanding metrics.

Students can start with the electronics they already know and progressively move toward Qiskit and quantum circuits Description: Classical-to-Quantum

Electronics concept	**Student already knows**	**Quantum counterpart**
Transistor	Electronic switching device	Qubit as quantum information unit
ON/OFF	0/1	Measurement gives 0 or 1
Voltage/current state	Physical representation of information	Quantum state/amplitude
Logic gate	AND, OR, NOT	X, Y, Z, H, CNOT gates
NOT gate	Inverts 0 ↔ 1	X gate performs quantum bit flip
Digital circuit	Combination of logic gates	Quantum circuit
Noise	Disturbs transistor signal	Decoherence/quantum noise
Measurement	Reading circuit output	Quantum measurement

Develop a **“Transistor-to-Qubit Learning Framework”** consisting of four levels.

Level 1 — Transistor

Begin with a simple transistor switching circuit.

For example:

Input → Transistor → LED

They experimentally observe: $V_{in} = 0 \rightarrow OFF$

$V_{in} = 1 \rightarrow ON$

Thus: $0/1 \rightarrow$ classical bit

Level 2 — Logic Gate

Construct a transistor-based NOT gate.

$$0 \rightarrow 1$$

$$1 \rightarrow 0$$

Then introduce the quantum equivalent:

$$X|0\rangle = |1\rangle$$

$$X|1\rangle = |0\rangle$$

X gate is analogous to a quantum NOT operation.

Level 3 — Bit → Qubit

Now introduce the key conceptual difference.

A classical bit is: $b \in \{0, 1\}$

A qubit can be represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where

$$|\alpha|^2 + |\beta|^2 = 1$$

experiment with the **Hadamard gate**:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

This produces:

$$P(0) = 50\%$$

$$P(1) = 50\%$$

This provides an important conceptual transition:

Transistor

> ON or OFF

Classical bit

> 0 or 1

Qubit

> Quantum state with probability amplitudes

Level 4 — Classical Circuit vs Quantum Circuit

Implement the same logical function using both approaches.

Classical

Transistors → Logic gates → Digital output

Quantum

Qubits → Quantum gates → Measurement

Practical Student Project

Project: Transistor-to-Qubit Simulator

Develop a small educational platform containing:

Module A — Transistor simulator

Students vary:

- Base/gate voltage
- Collector/drain current
- Switching threshold
- Load resistance

and observe transistor switching.

Module B — Logic-gate simulator

Implement:

- NOT
- AND
- OR
- NAND
- NOR
- XOR

Module C — Quantum simulator

Implement:

- $|0\rangle$
- $|1\rangle$
- X gate
- H gate
- Z gate
- CNOT
- Measurement

Module D — Comparison

The platform displays:

Classical	Quantum
Bit	Qubit
0/1	α

NOT	X
Logic operation	Quantum gate
Output	Measurement
Noise	Quantum noise
Deterministic	Probabilistic measurement

Technology Stack

Electronics

- BJT/MOSFET
- Breadboard
- Resistors
- LEDs
- Arduino/ESP32

Programming

- Python
- NumPy

Quantum computing:: Qiskit

- IBM Quantum Composer/simulator

First build the classical circuit physically and then reproduce the conceptual operation using a quantum simulator.

Once understand the basic mapping, introduce:

Transistor noise

$$V_{out} = V_{ideal} + n(t)$$

and compare it with quantum noise/decoherence.

Investigate:

> **How does classical noise in transistor-based switching compare conceptually with noise affecting quantum states?**

Then introduce: Bit flip error, Phase flip error, Depolarizing noise, Quantum error correction